

INTELLIGENT PHISHING URL DETECTION

Explainable AI (XAI) Security Analysis Report • Generated on 2026-08-20 03:54:02 UTC

TARGET URL: http://paypal-security-update.account-verify.xyz/signin.php?user=test	PREDICTION: PHISHING
Domain: paypal-security-update.account-verify.xyz	Phishing Probability: 75.0% (High Risk)
Model Architecture: Random Forest	Model Confidence: 75.0%

1. Explainable AI (SHAP Feature Attribution)

The chart below lists the top features that contributed most heavily towards or against the phishing classification:

Feature Name	Extracted Value	Contribution	Impact Direction
TLD Risk Factor	0.8	+0.228	+0.228 (Phishing Risk)
Subdomain Count	1	-0.131	-0.131 (Safe/Benign)
Suspicious Keywords Count	2	+0.106	+0.106 (Phishing Risk)
Path Length	21	-0.104	-0.104 (Safe/Benign)
HTTPS Protocol	0	+0.049	+0.049 (Phishing Risk)
Domain Length	41	+0.030	+0.030 (Phishing Risk)

2. Extracted URL Lexical & Heuristic Features

Feature Metric	Value	Feature Metric	Value
URL Length	69	Subdomain Count	1
Domain Length	41	Dot Count	3
HTTPS Encryption	No (Insecure)	Hyphen Count	3
Direct IP Host	No (Domain Name)	Digit Count	0
Suspicious Keywords	2	Shannon Entropy	4.41
Contains @ Symbol	No	TLD Risk Rating	80%

3. AI Cybersecurity Mitigation Recommendations

- {'category': 'Immediate User Action', 'priority': 'CRITICAL', 'icon': 'ShieldAlert', 'title': 'Do NOT Enter Credentials or Sensitive Information', 'action': 'Immediately close the browser tab. Under no circumstances should you input passwords, 2FA tokens, or financial details on this destination.', 'details': 'The URL displays active indicators of deceptive phishing targeting paypal-security-update.account-verify.xyz.'}
- {'category': 'Endpoint & Browser Defense', 'priority': 'HIGH', 'icon': 'Lock', 'title': 'Isolate Active Sessions & Reset Potentially Compromised Accounts', 'action': 'If you previously entered credentials on this site, immediately navigate to the legitimate official website directly (by manually typing the verified domain) and update your password. Enable multi-factor authentication (MFA/FIDO2).', 'details': 'Credential harvesters typically attempt automated login within seconds of submission.'}
- {'category': 'SOC & Enterprise Containment', 'priority': 'HIGH', 'icon': 'Server', 'title': 'Enforce DNS Sinkholing & Perimeter Firewall Block', 'action': 'Block outbound traffic to domain 'paypal-security-update.account-verify.xyz' across corporate firewalls, secure web

gateways (SWG), and DNS resolvers.", 'details': 'Domain IOC: paypal-security-update.account-verify.xyz. Recommend logging all recent egress DNS lookups to audit impacted endpoints.'}

- {'category': 'Indicator Specific', 'priority': 'MEDIUM', 'icon': 'AlertTriangle', 'title': 'Unencrypted HTTP Protocol', 'action': 'Never transmit credentials, cookies, or session tokens over unencrypted HTTP. Data is exposed in cleartext to network eavesdroppers.', 'details': 'Ensure all sensitive web endpoints enforce HTTPS with valid TLS 1.3 encryption.'}
- {'category': 'Indicator Specific', 'priority': 'MEDIUM', 'icon': 'FileQuestion', 'title': 'Deceptive Subdomain / Brand Typosquatting Alert', 'action': 'Inspect the root domain carefully before trusting any branded logo on the web page. Attackers often prefix legitimate brand names to foreign root domains.', 'details': 'Check that the domain truly terminates at the verified brand suffix, not a deceptive multi-tier subdomain.'}